

Orchestration composition is a Zappa–Szép product: a worked instantiation, and re-entrancy as the H^2 obstruction

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Abstract

We ground the “agent orchestration = directed container” dictionary on the smallest non-trivial case. A supervisor–worker orchestration is modelled as a small category (equivalently a directed container, Ahman–Uustalu); composing two such orchestrations into a single interleaved joint agent is a *distributive law / Zappa–Szép product* $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$, governed by the pairwise criterion (L) $\wedge$ (G). We exhibit, and machine-check, a topology table with four regimes. Three *compose*: independent supervisors (trivial law, $\mathcal{K} = \mathcal{C} \times \mathcal{D}$); a genuinely non-abelian coherent interleaving ($\mathcal{K} = S_3 = \mathbb{Z}/3 \bowtie \mathbb{Z}/2$); and a state-protected (locked) re-entry pattern (a $\mathbb{Z}/2$ -torsor of joint agents). One is *obstructed*: an *unprotected re-entrancy* pattern, in which a worker’s outcome mutates the supervisor’s shared state, presents *exactly* as the ten-morphism rigid twist — (L) holds but (G) fails, so no distributive law exists, and the obstruction is the nonzero class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/2$. This is the “re-entrancy = failed distributive law” reading, computed rather than asserted. The single bit that flips the verdict is whether a callee mutates the caller’s shared state. All claims are machine-checked (`projects/scratch/orchestration.zs*.py`); the load-bearing theorems are cited, not reproved.

1 What is and is not claimed

The abstract machinery is established and used as a black box.

- (T1) *Directed container* $\cong$ *small category* (Ahman–Uustalu; formalised in Lean in the author’s development). A handoff topology — objects are roles/agents, morphisms are admissible handoff paths — is a small category, hence a directed container.
- (T2) *Pairwise Zappa–Szép criterion* [1]. For a small category $\mathcal{K}$ and a wide subcategory $\mathcal{D} \subseteq \mathcal{K}$, there is a wide $\mathcal{C}$ with $(\mathcal{C}, \mathcal{D})$ a strict factorization system $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ if and only if (L) every hom-presheaf $\mathrm{Hom}_{\mathcal{K}}(-, b)$ is free as a right $\mathcal{D}$ -set, and (G) free bases can be chosen closing into a wide subcategory. The data $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ is exactly a distributive law $\lambda : \mathcal{D}\mathcal{C} \Rightarrow \mathcal{C}\mathcal{D}$ satisfying the matched-pair axioms ZS1–ZS4.
- (T3) *The (G)-obstruction is an H^2 class* [2]. Under the standard hypothesis (abelian vertex groups, trivial left action), (G) $\Leftrightarrow [\omega] = 0 \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$; the rigid twist realizes the generator of $H^2 \cong \mathbb{Z}/2$. The cohomology itself is classical (Rosebrugh–Wood for existence, Baues–Wirsching for the abelian classification, Pirashvili for the non-abelian case) and is cited, never reproved.

The reading “an agent’s interface is a polynomial functor / container; its dynamics is a coalgebra” is Spivak’s prior art and is not claimed here. **The sole contribution of this note** is the

worked instantiation: that the supervisor–worker orchestration patterns below literally present as the categories to which (T1)–(T3) apply, turning the dictionary from a proposed reading into a computed one. In particular the models are minimal faithful abstractions; we do not claim any specific production framework is *exactly* one of them.

2 The orchestration category

Fix the handoff roles S (supervisor), W (a worker running), R (a result/return point). The supervisor carries a one-bit *turn token* recording which worker currently holds its attention; passing the token is an involution τ , $\tau^2 = \text{id}$, so $\text{End}(S) \supseteq \{\text{id}_S, \tau\} \cong \mathbb{Z}/2$. The generating handoffs are

$$p, p\tau : S \rightarrow W \quad (\text{dispatch, in turn-state 0 resp. 1; } p\tau = p \circ \tau), \quad s, s_2 : W \rightarrow R \quad (\text{two worker outcomes}),$$

with composites $q, q\tau : S \rightarrow R$. The right $\mathbb{Z}/2$ -action is $p \cdot \tau = p\tau$, $q \cdot \tau = q\tau$. We take the prescribed right factor to be the supervisor-internal (“token/reader”) moves

$$\mathcal{D} = \{\text{id}_S, \tau, \text{id}_W, \text{id}_R\},$$

a wide subcategory. The candidate left factor $\mathcal{C}$ we are solving for is the genuine dispatch/handoff part. Everything hinges on *one* composite: how a worker outcome, run after a dispatch, sits relative to the token.

Definition 1 (The two regimes). Write s for “the worker returns normally” and s_2 for “the worker re-enters” (calls back into the supervisor before returning).

- (bug) *Unprotected re-entry*. The re-entrant branch flips the token: $s \circ p = q$ but $s_2 \circ p = q\tau$ (forcing $s \circ p\tau = q\tau$, $s_2 \circ p\tau = q$). The category is $\mathcal{K}_{\text{bug}}$.
- (ok) *State-protected re-entry (a lock)*. Neither outcome touches the token: $s \circ p = s_2 \circ p = q$, $s \circ p\tau = s_2 \circ p\tau = q\tau$. The category is $\mathcal{K}_{\text{ok}}$.

Both $\mathcal{K}_{\text{bug}}, \mathcal{K}_{\text{ok}}$ are genuine categories and $\mathcal{D}$ is wide in each (machine-checked). The reading is exact: *re-entrancy is the callee mutating shared caller state during a pending call*, and here that is precisely the s_2 -branch dragging the supervisor’s turn token, i.e. $s_2 \circ p = q\tau \neq q$.

Remark 2 (Relation to the free category on $S \rightleftarrows \{w_1, w_2\}$). This is the minimal finite faithful model of the two-worker re-dispatch pattern. The two worker outcomes s, s_2 are the two-worker choice “re-dispatch to w_1 vs. w_2 ”, and the involution τ is the token’s monodromy around the cycle $w_1 \rightarrow S \rightarrow w_2 \rightarrow S \rightarrow w_1$: traversing it once toggles the token, so the loop carries a $\mathbb{Z}/2$ holonomy. The free category on the graph $S \rightleftarrows \{w_1, w_2\}$ is infinite (its morphisms are all dispatch/return paths) and carries no relations, hence no finite obstruction; the obstruction is created by the *shared-state* relation $s_2 \circ p = q\tau$, which is exactly the identification a re-entrant, state-mutating orchestration imposes. We work with the finite quotient because the $\mathbb{Z}/2$ holonomy is a finite phenomenon the Zappa–Szép machinery computes directly.

3 The obstruction: unprotected re-entry is the rigid twist

Proposition 3 ($\mathcal{K}_{\text{bug}}$ is obstructed). *In $\mathcal{K}_{\text{bug}}$, (L) holds (each $\text{Hom}(-, b)$ is free over $\mathcal{D}$, with exactly two bases per target), but (G) fails: no wide $\mathcal{C}$ makes $(\mathcal{C}, \mathcal{D})$ a strict factorization system. Hence no distributive law $\lambda : \mathcal{D}\mathcal{C} \Rightarrow \mathcal{C}\mathcal{D}$ exists, and the two concurrent behaviours (the supervisor’s dispatch and the worker’s internal step) cannot be serialized into a single joint agent.*

Proof. Freeness is a direct check; we give the crux. The right $\mathcal{D}$ -orbits in $\text{Hom}(-, \mathbf{R})$ are $\{q, q\tau\}$ (source $\mathbf{S}$), $\{s\}$, $\{s_2\}$ (source $\mathbf{W}$; $\mathcal{D}$ has no non-identity arrow out of $\mathbf{W}$), and $\{\text{id}_{\mathbf{R}}\}$; a free basis picks one of $q, q\tau$ together with $s, s_2, \text{id}_{\mathbf{R}}$ — two bases. In $\text{Hom}(-, \mathbf{W})$ the orbits are $\{p, p\tau\}$ and $\{\text{id}_{\mathbf{W}}\}$; a basis is $\{\text{id}_{\mathbf{W}}, p\}$ or $\{\text{id}_{\mathbf{W}}, p\tau\}$. So (L) holds. Now any closing $\mathcal{C}$ chooses a $\mathbf{W}$ -basis element $t \in \{p, p\tau\}$ and an $\mathbf{R}$ -basis element $r \in \{q, q\tau\}$. Closure requires $s \circ t, s_2 \circ t \in \mathcal{C}$, i.e. both must be the chosen r . But $s \circ p = q$ and $s_2 \circ p = q\tau$ are the *two distinct* orbit representatives, so the $\mathbf{R}$ -basis would have to contain both q and $q\tau$ — impossible, it contains exactly one. The case $t = p\tau$ ($s \circ p\tau = q\tau$, $s_2 \circ p\tau = q$) is identical. Hence (G) fails; by (T2) there is no strict factorization, equivalently no distributive law. A brute-force search over all wide subcategories confirms $\#\text{SFS} = 0$. $\square$

Theorem 4 (The class is the generator of $\mathbb{Z}/2$). $\mathcal{K}_{\text{bug}}$ is isomorphic, as a category with its distinguished wide subcategory $\mathcal{D}$, to the ten-morphism rigid twist of [2] with $\mathcal{D} = \text{End}$, under

$$\mathbf{S} \mapsto a, \mathbf{W} \mapsto x, \mathbf{R} \mapsto y, \quad \tau \mapsto g, p \mapsto p, p\tau \mapsto pg, s \mapsto s, s_2 \mapsto s_2, q \mapsto q, q\tau \mapsto qg.$$

Consequently the hypothesis of (T3) holds ($\mathcal{D}$ is the abelian vertex group $\mathbb{Z}/2$ at $\mathbf{S}$, trivial elsewhere; the left action is trivial), the orbit category $\text{Sk}_{\mathcal{C}}$ is the branch $\mathbf{S} \rightarrow \mathbf{W} \rightrightarrows \mathbf{R}$, and

$$H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/2, \quad [\omega] \neq 0 \text{ (the generator)}.$$

Proof. The stated bijection is checked to preserve domains, codomains, all composites, and $\mathcal{D}$ (machine-verified in `orchestration_zs2.py`); it is therefore an isomorphism of categories matching the wide subcategories. The H^2 computation is then Theorem rt of [2] verbatim: normalized cochains give $Z^2 = C^2 \cong (\mathbb{Z}/2)^2$ (coordinates $\omega([s], [p]), \omega([s_2], [p])$), $B^2 = \langle (1, 1) \rangle$ (the diagonal), so $H^2 \cong \mathbb{Z}/2$, and the defect of the natural transversal is $(0, 1) \notin B^2$, the generator. $\square$

Remark 5 (Reading the class). The two coordinates of ω are the two worker outcomes; $B^2 =$ diagonal is the freedom to relabel which token-state is “canonical.” The class is nonzero precisely because the *two* outcomes disagree on that relabelling by one token-flip — the re-entrant branch’s side effect. This is the categorical avatar of “a flat bundle is trivial iff its holonomy vanishes”: the local completions of each partial handoff (free, by (L)) fail to glue into a global section (a serialized joint agent) by a single $\mathbb{Z}/2$ monodromy. It is the same obstruction the SEED reads in Ethereum re-entrancy, now located as an explicit H^2 generator on a handoff category.

4 The composing regimes

Proposition 6 (Locking restores composability). In $\mathcal{K}_{\text{ok}}$, (L) and (G) both hold; there are exactly two strict factorization systems over $\mathcal{D}$ (a $\mathbb{Z}/2$ -torsor, so $[\omega] = 0$), with $\mathcal{C} = \{\text{id}_{\mathbf{S}}, \text{id}_{\mathbf{W}}, \text{id}_{\mathbf{R}}, p, q, s, s_2\}$ in one, and the extracted distributive law λ satisfies ZS1–ZS4 and the unit laws.

Proof. Machine-checked (`orchestration_zs.py`, `orchestration_zs2.py`). The only change from $\mathcal{K}_{\text{bug}}$ is that $s_2 \circ p = q$ rather than $q\tau$, so $s \circ t$ and $s_2 \circ t$ now land on the *same* orbit representative for a fixed $\mathbf{W}$ -basis choice; the contradiction in the proof of Proposition 3 disappears and both transversals close. Extracting $\lambda(d, c)$ as the unique factorization of $d \circ c$ and checking ZS1–ZS4 returns no violations. $\square$

Two further composing regimes complete the table.

Example 7 (Independent supervisors). Two supervisors over read-only workers do not interfere: the distributive law is the swap $\lambda(d, c) = (c, d)$, which trivially satisfies ZS1–ZS4, and $\mathcal{K} = \mathcal{C} \times \mathcal{D}$. With $\mathcal{C} = \mathcal{D} = \mathbb{Z}/2$ (dispatch cycles truncated for finiteness) one gets $\mathcal{K} = \mathbb{Z}/2 \times \mathbb{Z}/2$, a category on the nose (checked).

Example 8 (Genuinely non-abelian coherent interleaving). Two supervisors over a *shared* worker pool can interleave nontrivially yet consistently. Take $\mathcal{C} = \mathbb{Z}/3$ (supervisor- $\mathcal{C}$ ’s cyclic worker rotation), $\mathcal{D} = \mathbb{Z}/2$ (supervisor- $\mathcal{D}$ ’s toggle), with the left action ${}^d c = -c$ for the non-identity d (the toggle reverses the rotation) and trivial right action. This is the dihedral matched pair; ZS1–ZS4 hold and $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D} = S_3$, with the join genuinely non-abelian $((c_1) \circ (d_1) \neq (d_1) \circ (c_1))$, checked). So “composition = Zappa–Szép product” is non-vacuous: it produces a legitimate joint agent that is neither a product nor obstructed. (The two-supervisor reading here is illustrative/schematic.)

5 The table

All four rows are machine-checked in `projects/scratch/orchestration.zs*.py`.

shared-worker topology	verdict	invariant
independent supervisors (read-only workers)	composes	$\mathcal{K} = \mathcal{C} \times \mathcal{D}$ (trivial law)
two supervisors, coherent nontrivial interleave	composes	$\mathcal{K} = S_3$, non-abelian join
re-entry, state-protected (lock)	composes	$\#SFS = 2$, $[\omega] = 0$
re-entry, unprotected (worker mutates state)	obstructed	$\#SFS = 0$, $[\omega] = \text{gen } \mathbb{Z}/2$

The single bit that flips the verdict is whether a worker’s outcome mutates the supervisor’s shared state. If it does, the distributive law becomes multivalued, (G) fails, and the obstruction to a joint agent is the nonzero $[\omega] \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$.

6 Grant relevance and honest status

This is the worked instantiation the orchestration dictionary lacked. It grounds the one genuinely unstaked claim — *composing two agent orchestrations is a distributive law / Zappa–Szép product* $\mathcal{C} \bowtie \mathcal{D}$, *not merely a functor* — on a computed example, and locates a concrete failure mode (unprotected re-entrancy) as an explicit degree-two cohomology class on the handoff category. The degree axis is the differentiator from the multi-agent-systems sheaf-Laplacian literature, which puts H^0 (consensus) and H^1 (identifiability) on the *communication* graph — *state* questions — whereas this puts H^2 on the *handoff* category — a *structural composability* question, one degree up.

Status. Grade: **computed**. The scaffolding (T1)–(T3) is proved and cited; the instantiation and the table are new and machine-checked; no new cohomology is claimed, and the models are minimal faithful abstractions rather than literal encodings of any named framework.

References

- [1] MacBeth, *The pairwise Zappa–Szép criterion*, Kodamai note, 10 June 2026.
- [2] MacBeth, *The global-closure obstruction (G) is a cohomology class: the Zappa–Szép holonomy as $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$* , Kodamai note, 11 June 2026.